

# GEN211 Probability and Statistics

## Lecture 2: Descriptive Statistics II: Data Presentation (cont.) and Data Summarization





# Lecture contents

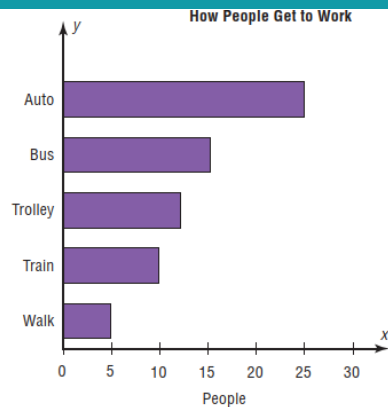
- Data presentation (cont.): Other types of graphs
- Data summarization I: Measure of central tendency
- Data summarization II: Measure of variation



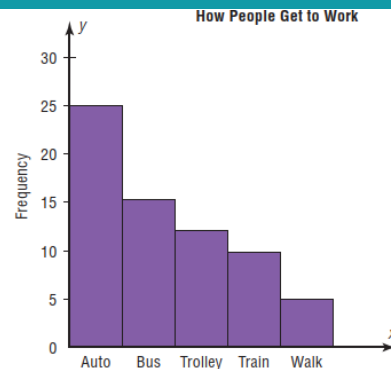
# Data Presentation (cont.): Other Types of Graphs



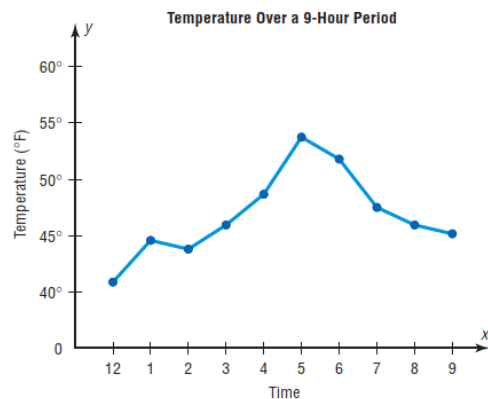
# Other types of graphs



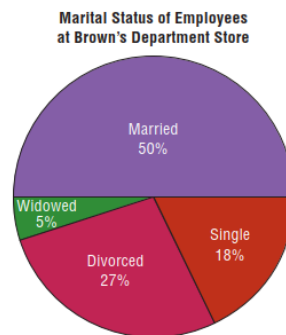
(a) Bar graph



(b) Pareto chart



(c) Time series graph



(d) Pie graph



# Example 1: Bar charts

A **bar graph** represents the data by using vertical or horizontal bars whose heights or lengths represent the frequencies of the data.

The table shows the average money spent by first-year college students. Draw a horizontal and vertical bar graph for the data.

|             |       |
|-------------|-------|
| Electronics | \$728 |
| Dorm decor  | 344   |
| Clothing    | 141   |
| Shoes       | 72    |

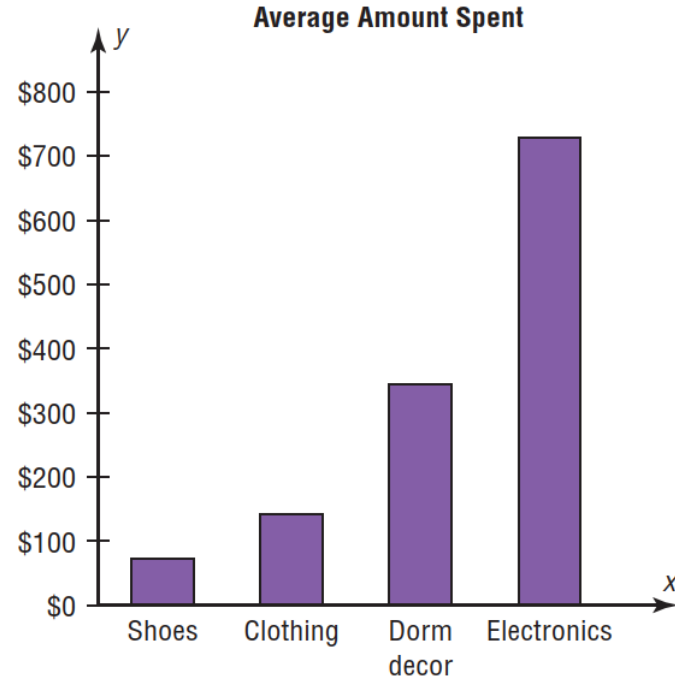
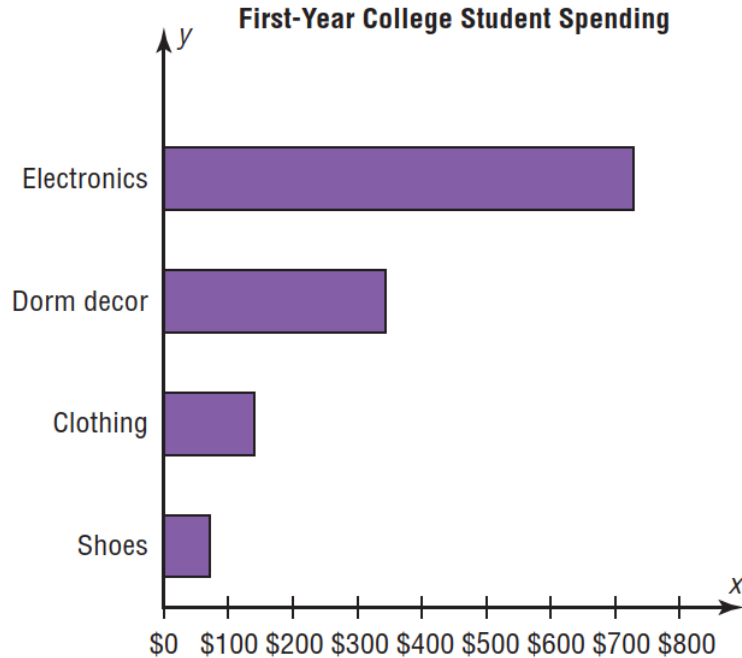
## SOLUTION

1. Draw and label the  $x$  and  $y$  axes. For the horizontal bar graph place the frequency scale on the  $x$  axis, and for the vertical bar graph place the frequency scale on the  $y$  axis.
2. Draw the bars corresponding to the frequencies.



# Example 1: Bar charts

## SOLUTION



## Example 2: Compound bar charts

Consider the following data for the number (in millions) of never married adults in the United States.

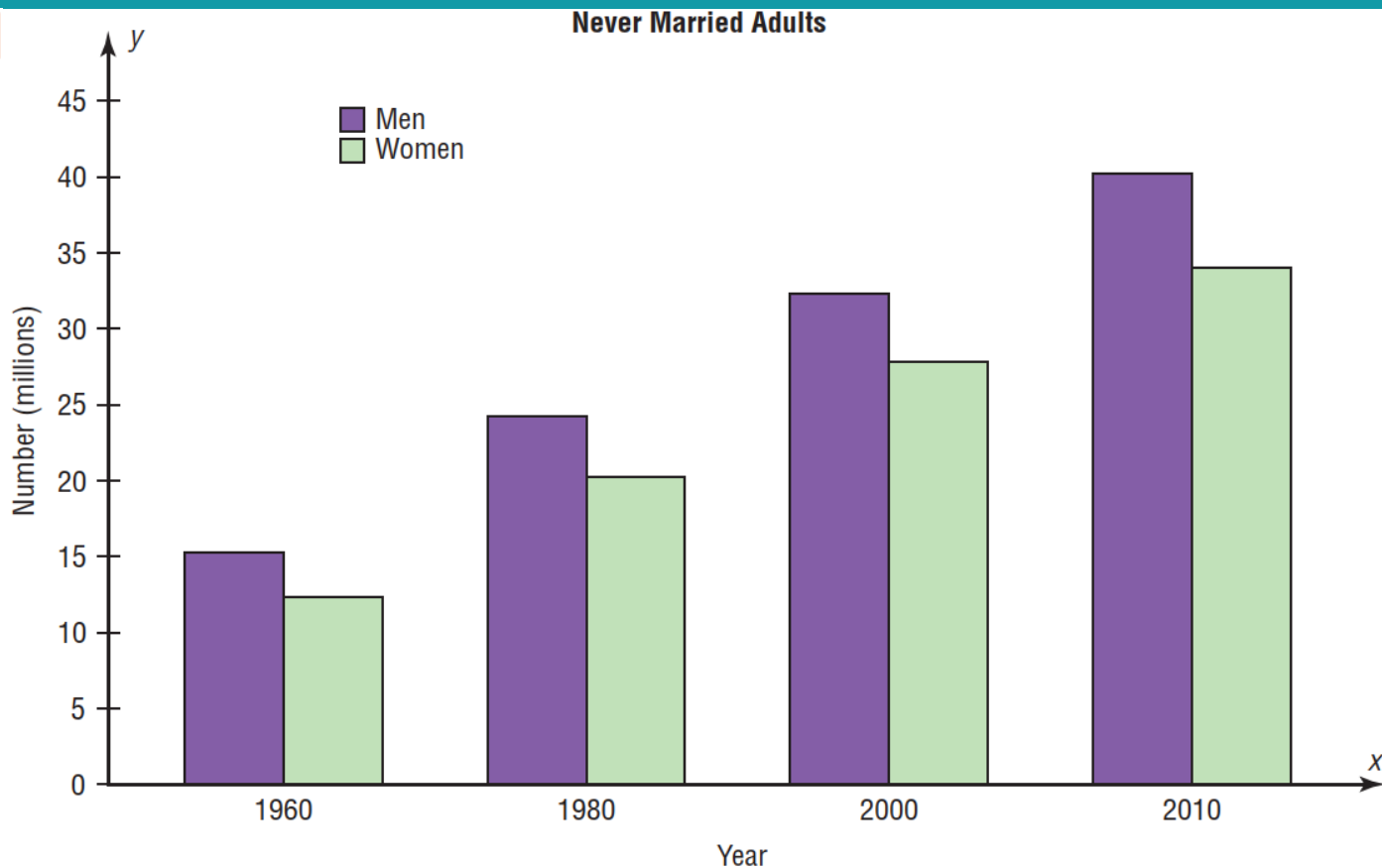
| Year | Males | Females |
|------|-------|---------|
| 1960 | 15.3  | 12.3    |
| 1980 | 24.2  | 20.2    |
| 2000 | 32.3  | 27.8    |
| 2010 | 40.2  | 34.0    |

- There have consistently been more never married males than never married females.
- The difference in the two groups has increased slightly over the last 50 years.



# Example 2: Compound bar charts

**SOLUTION**





# Example 3: Pareto charts

A **Pareto chart** is used to represent a frequency distribution for a categorical variable, and the frequencies are displayed by the heights of vertical bars, which are arranged in order from highest to lowest.

## Suggestions for Drawing Pareto Charts

1. Make the bars the same width.
2. Arrange the data from largest to smallest according to frequency.
3. Make the units that are used for the frequency equal in size.

The data shown here consist of the number of police calls for specific categories that a local municipality received in 2011. (The data were obtained by the author.) Draw and analyze a Pareto chart for the data.

| Category                    | Number |
|-----------------------------|--------|
| Juvenile complaint          | 92     |
| Loud noise/music/party      | 27     |
| Drug offenses               | 79     |
| Driving under the influence | 38     |
| Disabled vehicle            | 65     |



# Example 3: Pareto charts

## SOLUTION

**Step 1** Arrange the data from the largest to smallest according to frequency.

| Category                    | Number |
|-----------------------------|--------|
| Juvenile complaint          | 92     |
| Drug offenses               | 79     |
| Disabled vehicle            | 65     |
| Driving under the influence | 38     |
| Loud noise/music/party      | 27     |

**Step 2** Draw and label the  $x$  and  $y$  axes.

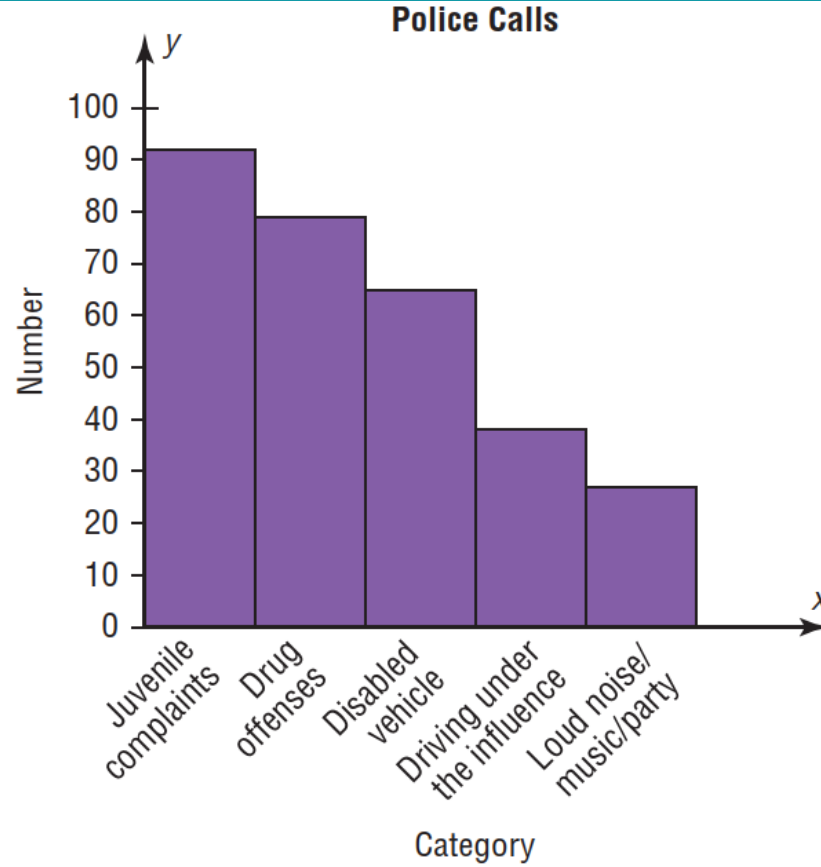
**Step 3** Draw the bars corresponding to the frequencies.

The graph shows that the police department received the most calls for juvenile complaints and the fewest calls for loud noise, music, and parties.



# Example 3: Pareto charts

## SOLUTION



## Example 4: The time series graph

A **time series graph** represents data that occur over a specific period of time.

The data show the percentage of U.S. adults who smoke. Draw and analyze a time series graph for the data.

| Year    | 1970 | 1980 | 1990 | 2000 | 2010 |
|---------|------|------|------|------|------|
| Percent | 37   | 33   | 25   | 23   | 19   |

### SOLUTION

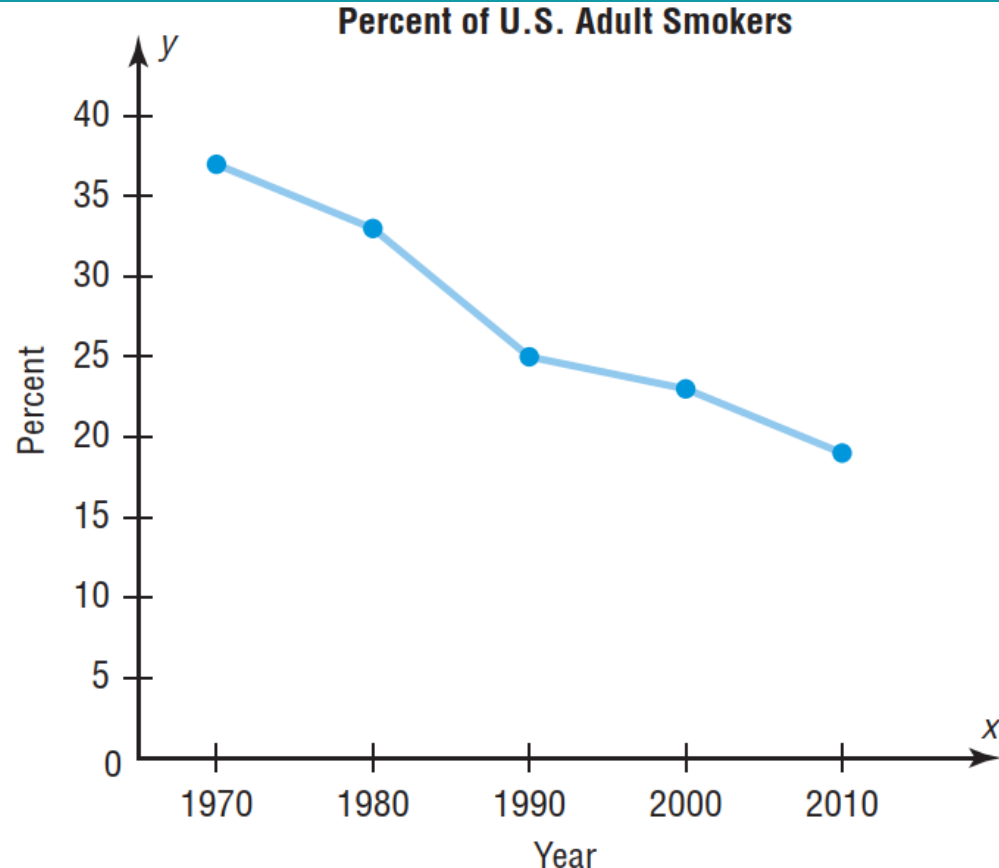
- Step 1** Draw and label the  $x$  and  $y$  axes.
- Step 2** Label the  $x$  axis for years and the  $y$  axis for the percent.
- Step 3** Plot each point according to the table.
- Step 4** Draw line segments connecting adjacent points. Do not try to fit a smooth curve through the data points.

There has been a decrease in the percent of smokers over the 40-year period.



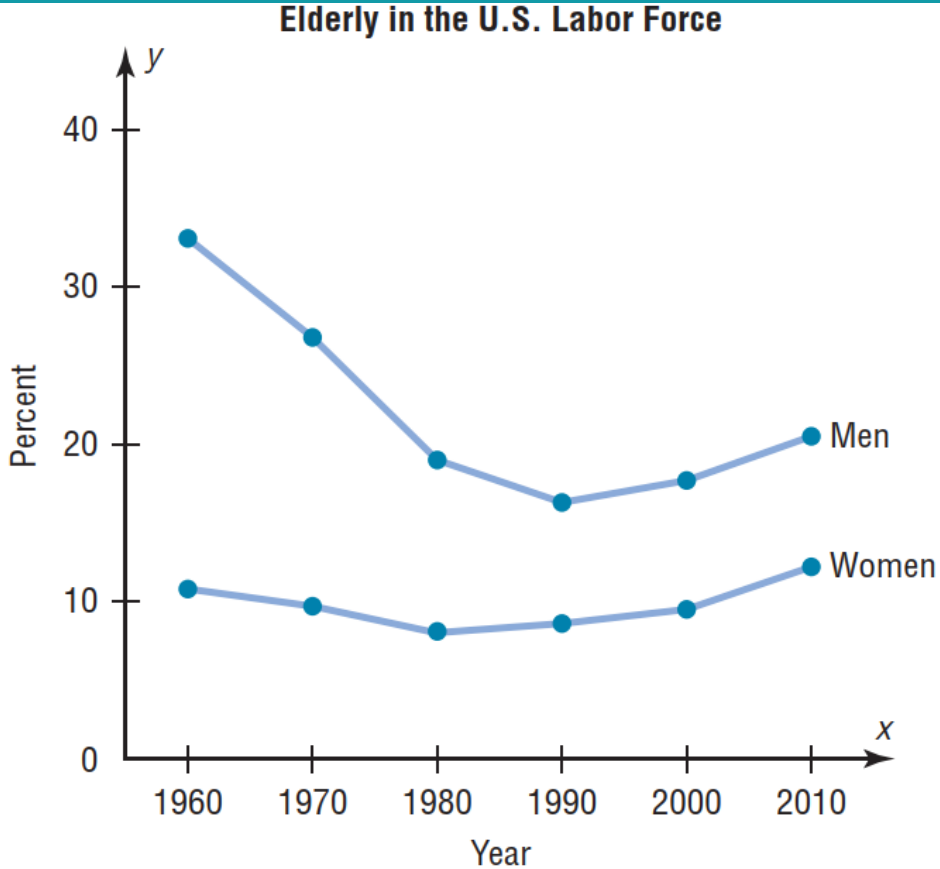
# Example 4: The time series graph

**SOLUTION**



# Compound time series graph

- This graph shows the percentage of elderly males and females in the U.S. labor force from 1960 to 2010.
- It shows that the percentage of elderly men decreased significantly from 1960 to 1990 and then increased slightly after that.
- For the elderly females, the percentage decreased slightly from 1960 to 1980 and then increased from 1980 to 2010.



## Example 5: The pie graph

A **pie graph** is a circle that is divided into sections or wedges according to the percentage of frequencies in each category of the distribution.

This frequency distribution shows the number of pounds of each snack food eaten during the Super Bowl. Construct a pie graph for the data.

| Snack                    | Pounds (frequency) |
|--------------------------|--------------------|
| Potato chips             | 11.2 million       |
| Tortilla chips           | 8.2 million        |
| Pretzels                 | 4.3 million        |
| Popcorn                  | 3.8 million        |
| Snack nuts               | 2.5 million        |
| Total $n = 30.0$ million |                    |



# Example 5: The pie graph

## SOLUTION

**Step 1** Since there are  $360^\circ$  in a circle, the frequency for each class must be converted to a proportional part of the circle. This conversion is done by using the formula

$$\text{Degrees} = \frac{f}{n} \cdot 360^\circ$$

where  $f$  = frequency for each class and  $n$  = sum of the frequencies. Hence, the following conversions are obtained. The degrees should sum to  $360^\circ$ .

|                |                                               |
|----------------|-----------------------------------------------|
| Potato chips   | $\frac{11.2}{30} \cdot 360^\circ = 134^\circ$ |
| Tortilla chips | $\frac{8.2}{30} \cdot 360^\circ = 98^\circ$   |
| Pretzels       | $\frac{4.3}{30} \cdot 360^\circ = 52^\circ$   |
| Popcorn        | $\frac{3.8}{30} \cdot 360^\circ = 46^\circ$   |
| Snack nuts     | $\frac{2.5}{30} \cdot 360^\circ = 30^\circ$   |
| Total          | <hr/> 360°                                    |





# Example 5: The pie graph

## SOLUTION

**Step 2** Each frequency must also be converted to a percentage.

$$\% = \frac{f}{n} \cdot 100$$

Hence, the following percentages are obtained. The percentages should sum to 100%. *Note:* The percent column does not always sum to 100% due to rounding.

$$\text{Potato chips} \quad \frac{11.2}{30} \cdot 100 = 37.3\%$$

$$\text{Tortilla chips} \quad \frac{8.2}{30} \cdot 100 = 27.3\%$$

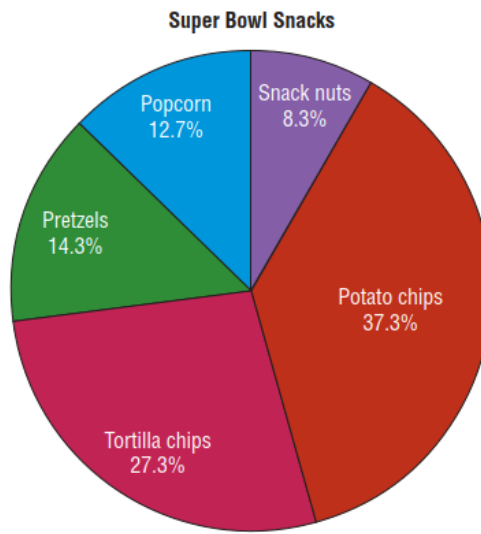
$$\text{Pretzels} \quad \frac{4.3}{30} \cdot 100 = 14.3\%$$

$$\text{Popcorn} \quad \frac{3.8}{30} \cdot 100 = 12.7\%$$

$$\text{Snack nuts} \quad \frac{2.5}{30} \cdot 100 = 8.3\%$$

$$\text{Total} \quad \underline{99.9\%}$$

**Step 3** Next, using a protractor and a compass, draw the graph, using the appropriate degree measures found in Step 1, and label each section with the name and percentages.



## Example 6: The pie graph

Construct and analyze a pie graph for the calls received each shift by a local municipality for 2011. (Data obtained by author.)

| Shift      | Frequency |
|------------|-----------|
| 1. Day     | 2594      |
| 2. Evening | 2800      |
| 3. Night   | 2436      |
|            | <hr/>     |
|            | 7830      |



## Example 6: The pie graph

### SOLUTION

**Step 1** Find the number of degrees for each shift, using the formula:

$$\text{Degrees} = \frac{f}{n} \cdot 360^\circ$$

For each shift, the following results are obtained:

$$\text{Day: } \frac{2594}{7830} \cdot 360^\circ = 119^\circ$$

$$\text{Evening: } \frac{2800}{7830} \cdot 360^\circ = 129^\circ$$

$$\text{Night: } \frac{2436}{7830} \cdot 360^\circ = 112^\circ$$



## Example 6: The pie graph

### SOLUTION

**Step 2** Find the percentages:

$$\text{Day: } \frac{2594}{7830} \cdot 100 = 33\%$$

$$\text{Evening: } \frac{2800}{7830} \cdot 100 = 36\%$$

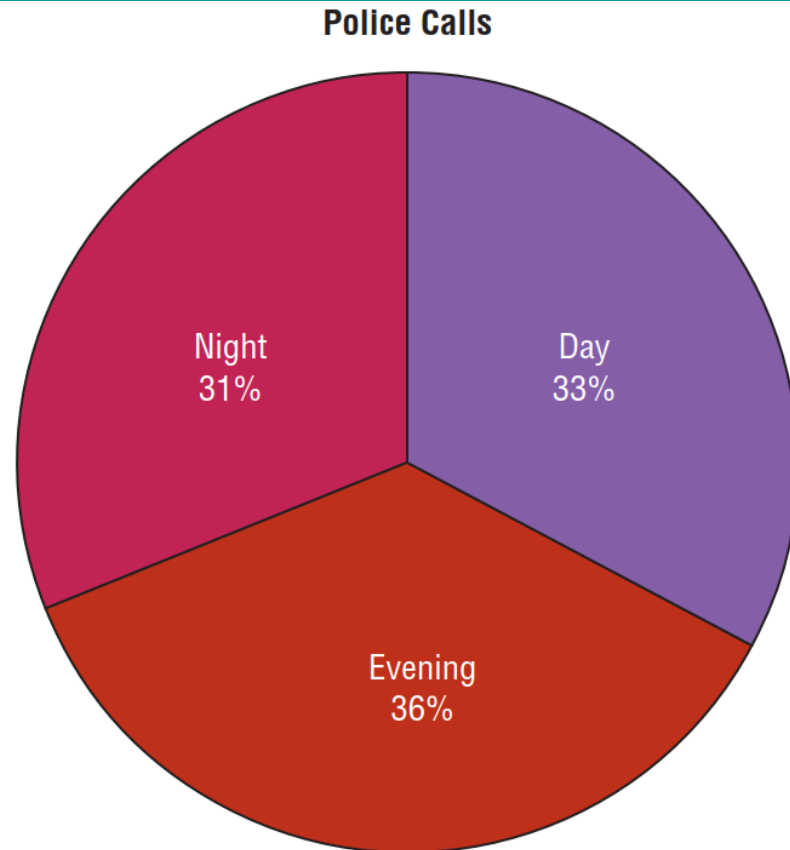
$$\text{Night: } \frac{2436}{7830} \cdot 100 = 31\%$$

**Step 3** Using a protractor, graph each section and write its name and corresponding percentage.



# Example 6: The pie graph

**SOLUTION**



# Data Summarization I: Measure of Central Tendency



# Statistics vs. parameters

A **statistic** is a characteristic or measure obtained by using the data values from a sample.

A **parameter** is a characteristic or measure obtained by using all the data values from a specific population.

Measures found by using all the data values in the population are called *parameters*. Measures obtained by using the data values from samples are called *statistics*; hence, the average of the sales from a sample of representatives is a *statistic*, and the average of sales obtained from the entire population is a *parameter*.



# Sample vs. population means

The **mean** is the sum of the values, divided by the total number of values.

The **sample mean**, denoted by  $\bar{X}$  (pronounced “X bar”), is calculated by using sample data. The sample mean is a statistic.

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} = \frac{\Sigma X}{n}$$

where  $n$  represents the total number of values in the sample.

The **population mean**, denoted by  $\mu$  (pronounced “mew”), is calculated by using all the values in the population. The population mean is a parameter.

$$\mu = \frac{X_1 + X_2 + X_3 + \dots + X_N}{N} = \frac{\Sigma X}{N}$$

where  $N$  represents the total number of values in the population.

The mean, in most cases, is not an actual data value.





## Examples 7, 8 on the sample mean

The number of calls that a local police department responded to for a sample of 9 months is shown. Find the mean.

475, 447, 440, 761, 993, 1052, 783, 671, 621

### SOLUTION

$$\begin{aligned}\bar{X} &= \frac{\sum x}{n} = \frac{475 + 447 + 440 + 761 + 993 + 1052 + 783 + 671 + 621}{9} \\ &= \frac{6243}{9} \approx 693.7\end{aligned}$$

Hence, the mean number of incidents per month to which the police responded is 693.7.

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The data show the number of patients in a sample of six hospitals who acquired an infection while hospitalized. Find the mean.

110      76      29      38      105      31

### SOLUTION

$$\bar{X} = \frac{\sum X}{n} = \frac{110 + 76 + 29 + 38 + 105 + 31}{6} = \frac{389}{6} = 64.8$$

The mean of the number of hospital infections for the six hospitals is 64.8.



# Mean for grouped data

## Procedure Table

### Finding the Mean for Grouped Data

**Step 1** Make a table as shown.

| A     | B             | C              | D             |
|-------|---------------|----------------|---------------|
| Class | Frequency $f$ | Midpoint $X_m$ | $f \cdot X_m$ |

**Step 2** Find the midpoints of each class and place them in column C.

**Step 3** Multiply the frequency by the midpoint for each class, and place the product in column D.

**Step 4** Find the sum of column D.

**Step 5** Divide the sum obtained in column D by the sum of the frequencies obtained in column B.

The formula for the mean is

$$\bar{X} = \frac{\Sigma f \cdot X_m}{n}$$

[Note: The symbols  $\Sigma f \cdot X_m$  mean to find the sum of the product of the frequency ( $f$ ) and the midpoint ( $X_m$ ) for each class.]



## Example 9: Mean for grouped data

Using the following frequency distribution, find the mean.

The data represent the number of miles run during one week for a sample of 20 runners.

| Class boundaries | Frequency |
|------------------|-----------|
| 5.5–10.5         | 1         |
| 10.5–15.5        | 2         |
| 15.5–20.5        | 3         |
| 20.5–25.5        | 5         |
| 25.5–30.5        | 4         |
| 30.5–35.5        | 3         |
| 35.5–40.5        | 2         |
|                  | <u>20</u> |
|                  | Total     |



# Example 9: Mean for grouped data

## SOLUTION

**Step 1** Make a table as shown.

| A<br>Class | B<br>Frequency $f$  | C<br>Midpoint $X_m$ | D<br>$f \cdot X_m$ |
|------------|---------------------|---------------------|--------------------|
| 5.5–10.5   | 1                   |                     |                    |
| 10.5–15.5  | 2                   |                     |                    |
| 15.5–20.5  | 3                   |                     |                    |
| 20.5–25.5  | 5                   |                     |                    |
| 25.5–30.5  | 4                   |                     |                    |
| 30.5–35.5  | 3                   |                     |                    |
| 35.5–40.5  | 2                   |                     |                    |
|            | $n = \overline{20}$ |                     |                    |



# Example 9: Mean for grouped data

## SOLUTION

**Step 2** Find the midpoints of each class and enter them in column C.

$$X_m = \frac{5.5 + 10.5}{2} = 8 \quad \frac{10.5 + 15.5}{2} = 13 \quad \text{etc.}$$

**Step 3** For each class, multiply the frequency by the midpoint, as shown, and place the product in column D.

$$1 \cdot 8 = 8 \quad 2 \cdot 13 = 26 \quad \text{etc.}$$

The completed table is shown here.

| A<br>Class | B<br>Frequency $f$ | C<br>Midpoint $X_m$ | D<br>$f \cdot X_m$         |
|------------|--------------------|---------------------|----------------------------|
| 5.5–10.5   | 1                  | 8                   | 8                          |
| 10.5–15.5  | 2                  | 13                  | 26                         |
| 15.5–20.5  | 3                  | 18                  | 54                         |
| 20.5–25.5  | 5                  | 23                  | 115                        |
| 25.5–30.5  | 4                  | 28                  | 112                        |
| 30.5–35.5  | 3                  | 33                  | 99                         |
| 35.5–40.5  | 2                  | 38                  | 76                         |
|            | $n = 20$           |                     | $\Sigma f \cdot X_m = 490$ |

**Step 4** Find the sum of column D.

**Step 5** Divide the sum by  $n$  to get the mean.

$$\bar{X} = \frac{\Sigma f \cdot X_m}{n} = \frac{490}{20} = 24.5 \text{ miles}$$



# The median

The **median** is the midpoint of the data array. The symbol for the median is MD.

## Procedure Table

### Finding the Median

**Step 1** Arrange the data values in ascending order.

**Step 2** Determine the number of values in the data set.

**Step 3** *a.* If  $n$  is odd, select the middle data value as the median.

*b.* If  $n$  is even, find the mean of the two middle values. That is, add them and divide the sum by 2.



## Example 10: Odd number of values in the data set

The number of police officers killed in the line of duty over the last 11 years is shown. Find the median.

177 153 122 141 189 155 162 165 149 157 240

### SOLUTION

**Step 1** Arrange the data in ascending order.

122, 141, 149, 153, 155, 157, 162, 165, 177, 189, 240

**Step 2** There are an odd number of data values, namely, 11.

**Step 3** Select the middle data value.

122, 141, 149, 153, 155, 157, 162, 165, 177, 189, 240



Median

Page - 31 The median number of police officers killed for the 11-year period is 157.



## Example 11: Even number of values in the data set

The number of tornadoes that have occurred in the United States over an 8-year period follows. Find the median.

684, 764, 656, 702, 856, 1133, 1132, 1303

### SOLUTION

**Step 1** Arrange the data values in ascending order.

656, 684, 702, 764, 856, 1132, 1133, 1303

**Step 2** There are an even number of data values, namely, 8.

**Step 3** The middle two data values are 764 and 856.

656, 684, 702, 764, 856, 1132, 1133, 1303



Median

Since the middle point falls halfway between 764 and 856, find the median MD by adding the two values and dividing by 2.

$$\text{MD} = \frac{764 + 856}{2} = \frac{1620}{2} = 810$$

The median number of tornadoes is 810.





## Example 12: The mode (unimodal data set)

The value that occurs most often in a data set is called the **mode**.

Find the mode of the signing bonuses of eight NFL players for a specific year. The bonuses in millions of dollars are

18.0, 14.0, 34.5, 10, 11.3, 10, 12.4, 10

### SOLUTION

It is helpful to arrange the data in order, although it is not necessary.

10, 10, 10, 11.3, 12.4, 14.0, 18.0, 34.5

Since \$10 million occurred 3 times—a frequency larger than any other number—the mode is \$10 million.



## Examples 13, 14: The mode (bimodal data set and data sets with no mode)

The data show the number of licensed nuclear reactors in the United States for a recent 15-year period. Find the mode.

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 104 | 104 | 104 | 104 | 104 |
| 107 | 109 | 109 | 109 | 110 |
| 109 | 111 | 112 | 111 | 109 |

### **SOLUTION**

Since the values 104 and 109 both occur 5 times, the modes are 104 and 109. The data set is said to be bimodal.

The number of accidental deaths due to firearms for a six-year period is shown. Find the mode.

649, 789, 642, 613, 610, 600

### **SOLUTION**

Since each value occurs only once, there is no mode.



## Example 15: The modal class

The mode for grouped data is the modal class. The **modal class** is the class with the largest frequency.

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Find the modal class for the frequency distribution of miles that 20 runners ran in one week.

| Class     | Frequency       |
|-----------|-----------------|
| 5.5–10.5  | 1               |
| 10.5–15.5 | 2               |
| 15.5–20.5 | 3               |
| 20.5–25.5 | 5 ← Modal class |
| 25.5–30.5 | 4               |
| 30.5–35.5 | 3               |
| 35.5–40.5 | 2               |

### SOLUTION

The modal class is 20.5–25.5, since it has the largest frequency. Sometimes the midpoint of the class is used rather than the boundaries; hence, the mode could also be given as 23 miles per week.



## Example 16: Mode for a nominal or categorical data set

The data show the number of boats registered for six counties in southwestern Pennsylvania. Find the mode.

|              |        |
|--------------|--------|
| Westmoreland | 11,008 |
| Butler       | 9,002  |
| Washington   | 6,843  |
| Beaver       | 6,367  |
| Fayette      | 4,208  |
| Armstrong    | 3,782  |

### SOLUTION

Since the category with the highest frequency is Westmoreland, the most typical case is Westmoreland. Hence, the mode is Westmoreland.



## Example 17: Mean, median, and mode

A small company consists of the owner, the manager, the salesperson, and two technicians, all of whose annual salaries are listed here. (Assume that this is the entire population.)

| Staff       | Salary    |
|-------------|-----------|
| Owner       | \$100,000 |
| Manager     | 40,000    |
| Salesperson | 24,000    |
| Technician  | 18,000    |
| Technician  | 18,000    |

Find the mean, median, and mode.

### SOLUTION

$$\mu = \frac{\Sigma X}{N} = \frac{\$100,000 + 40,000 + 24,000 + 18,000 + 18,000}{5} = \frac{\$200,000}{5} = \$40,000$$

Hence, the mean is \$40,000, the median is \$24,000, and the mode is \$18,000.



## Example 18: The midrange

The **midrange** is defined as the sum of the lowest and highest values in the data set, divided by 2. The symbol MR is used for the midrange.

$$\text{MR} = \frac{\text{lowest value} + \text{highest value}}{2}$$

The number of bank failures for a recent five-year period is shown. Find the midrange.

3, 30, 148, 157, 71

### SOLUTION

The lowest data value is 3, and the highest data value is 157.

$$\text{MR} = \frac{3 + 157}{2} = \frac{160}{2} = 80$$

The midrange for the number of bank failures is 80.



## Example 19: The midrange

Find the midrange of data for the NFL signing bonuses. The bonuses in millions of dollars are

18.0, 14.0, 34.5, 10, 11.3, 10, 12.4, 10

### SOLUTION

The lowest bonus is \$10 million, and the largest bonus is \$34.5 million.

$$\text{MR} = \frac{10 + 34.5}{2} = \frac{44.5}{2} = \$22.25 \text{ million}$$

Notice that this amount is larger than seven of the eight amounts and is not typical of the average of the bonuses. The reason is that there is one very high bonus, namely, \$34.5 million.



# The weighted mean

Find the **weighted mean** of a variable  $X$  by multiplying each value by its corresponding weight and dividing the sum of the products by the sum of the weights.

$$\bar{X} = \frac{w_1X_1 + w_2X_2 + \cdots + w_nX_n}{w_1 + w_2 + \cdots + w_n} = \frac{\sum wX}{\sum w}$$

where  $w_1, w_2, \dots, w_n$  are the weights and  $X_1, X_2, \dots, X_n$  are the values.





## Example 20

A student received an A in English Composition I (3 credits), a C in Introduction to Psychology (3 credits), a B in Biology I (4 credits), and a D in Physical Education (2 credits). Assuming A = 4 grade points, B = 3 grade points, C = 2 grade points, D = 1 grade point, and F = 0 grade points, find the student's grade point average.

### SOLUTION

| Course                     | Credits ( $w$ ) | Grade ( $X$ ) |
|----------------------------|-----------------|---------------|
| English Composition I      | 3               | A (4 points)  |
| Introduction to Psychology | 3               | C (2 points)  |
| Biology I                  | 4               | B (3 points)  |
| Physical Education         | 2               | D (1 point)   |

$$\bar{X} = \frac{\sum wX}{\sum w} = \frac{3 \cdot 4 + 3 \cdot 2 + 4 \cdot 3 + 2 \cdot 1}{3 + 3 + 4 + 2} = \frac{32}{12} \approx 2.7$$



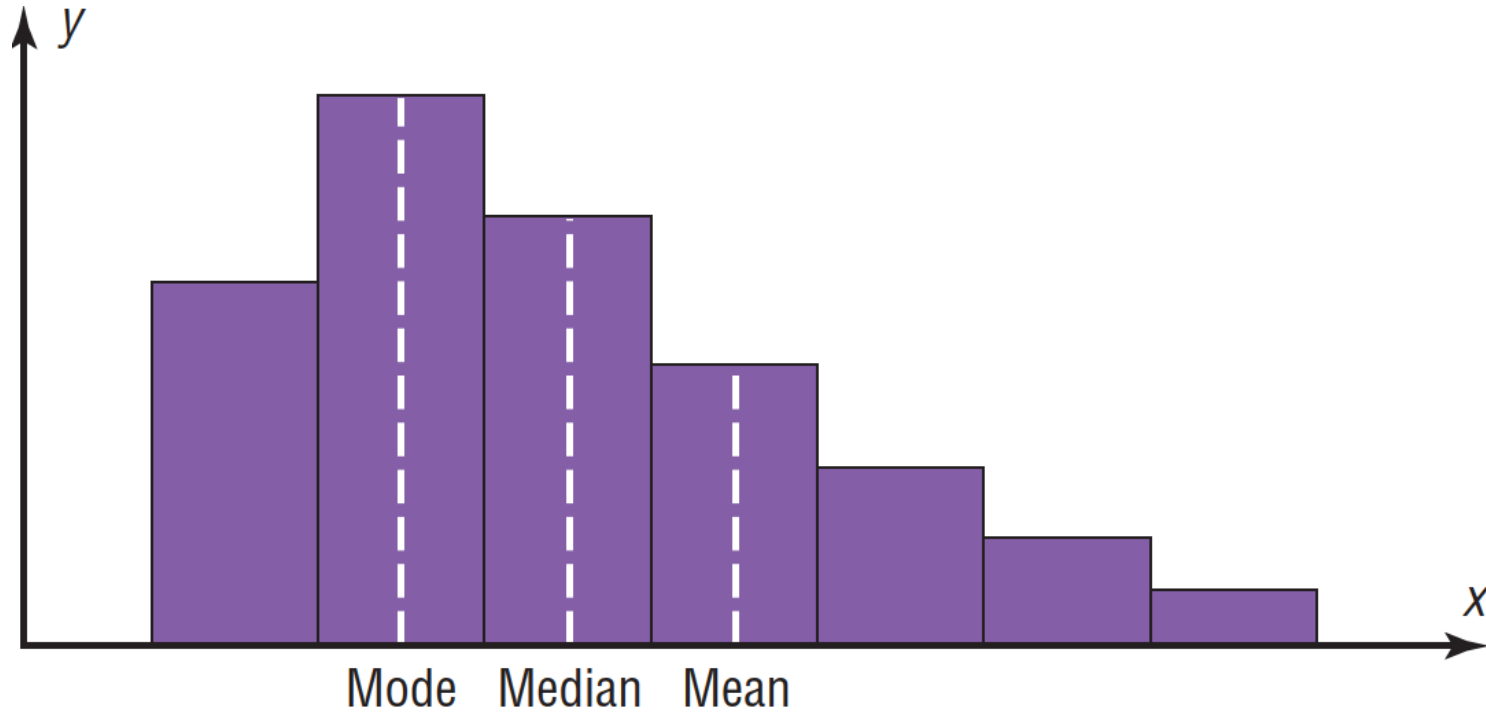
# Summary of measures of central tendency

**Summary of Measures of Central Tendency**

| Measure  | Definition                                       | Symbol(s)      |
|----------|--------------------------------------------------|----------------|
| Mean     | Sum of values, divided by total number of values | $\mu, \bar{X}$ |
| Median   | Middle point in data set that has been ordered   | MD             |
| Mode     | Most frequent data value                         | None           |
| Midrange | Lowest value plus highest value, divided by 2    | MR             |



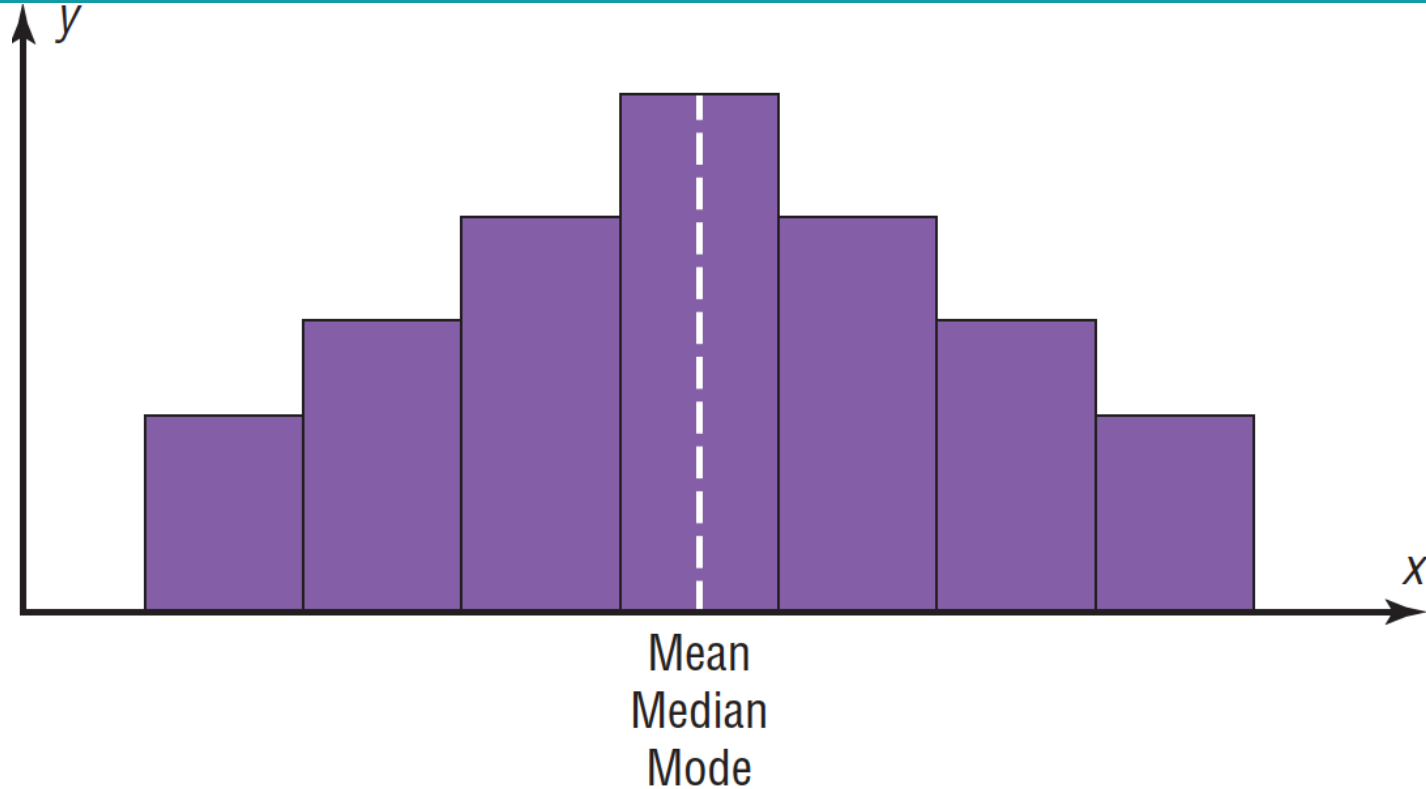
# Mean vs. median vs. mode for different distributions



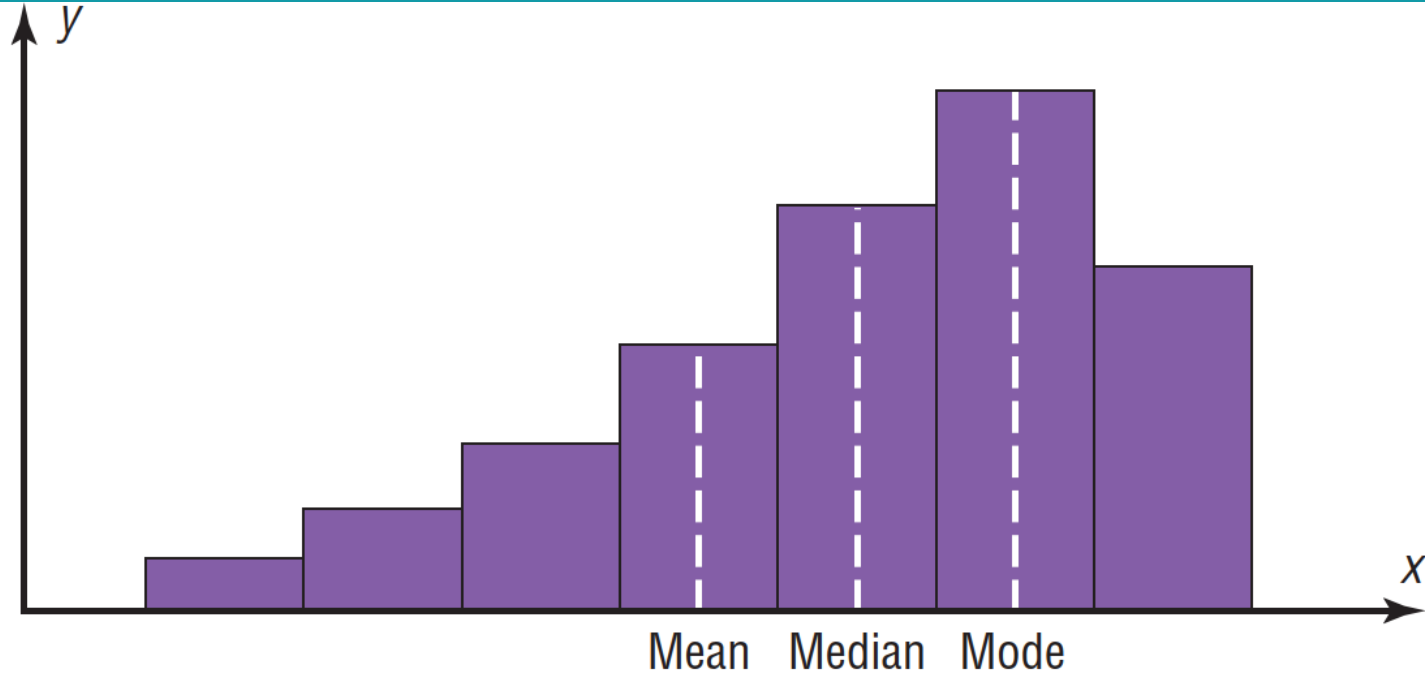
Page - 43 **(a)** Positively skewed or right-skewed



# Mean vs. median vs. mode for different distributions



# Mean vs. median vs. mode for different distributions



# Data Summarization II: Measure of Variation



# Motivation: Example on comparison of outdoor paint

A testing lab wishes to test two experimental brands of outdoor paint to see how long each will last before fading. The testing lab makes 6 gallons of each paint to test. Since different chemical agents are added to each group and only six cans are involved, these two groups constitute two small populations. The results (in months) are shown. Find the mean of each group.

| Brand A | Brand B |
|---------|---------|
| 10      | 35      |
| 60      | 45      |
| 50      | 30      |
| 30      | 35      |
| 40      | 40      |
| 20      | 25      |

The mean for brand A is

$$\mu = \frac{\sum X}{N} = \frac{210}{6} = 35 \text{ months}$$

The mean for brand B is

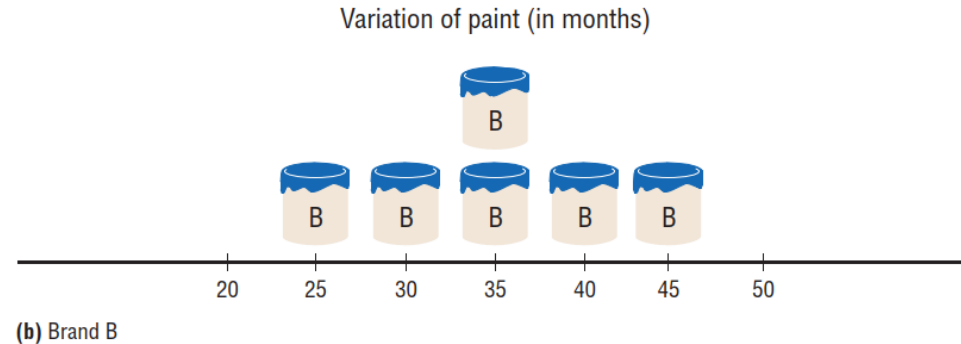
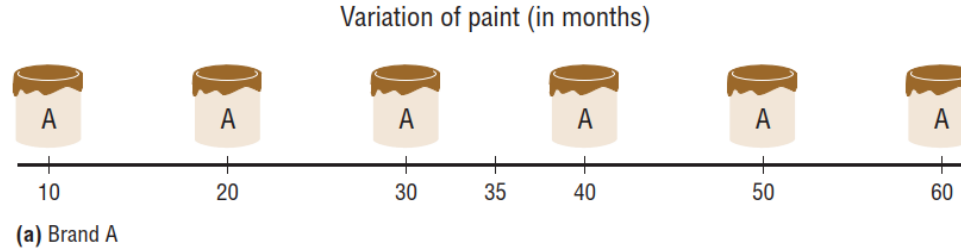
$$\mu = \frac{\sum X}{N} = \frac{210}{6} = 35 \text{ months}$$



# Motivation: Example on comparison of outdoor paint

Since the means are equal, you might conclude that both brands of paint last equally well. However, when the data sets are examined graphically, a somewhat different conclusion might be drawn.

even though the means are the same for both brands, the spread, or variation, is quite different.





## Example 21: The range

The **range** is the highest value minus the lowest value. The symbol  $R$  is used for the range.

$$R = \text{highest value} - \text{lowest value}$$

Find the ranges for the paints in the previous example, slides 47 and 48.

### SOLUTION

For brand A, the range is

$$R = 60 - 10 = 50 \text{ months}$$

For brand B, the range is

$$R = 45 - 25 = 20 \text{ months}$$

Make sure the range is given as a single number.

The range for brand A shows that 50 months separate the largest data value from the smallest data value. For brand B, 20 months separate the largest data value from the smallest data value, which is less than one-half of brand A's range.



## Example 22: The range

The salaries for the staff of the XYZ Manufacturing Co. are shown here. Find the range.

| Staff                | Salary    |
|----------------------|-----------|
| Owner                | \$100,000 |
| Manager              | 40,000    |
| Sales representative | 30,000    |
| Workers              | 25,000    |
|                      | 15,000    |
|                      | 18,000    |

### SOLUTION

The range is  $R = \$100,000 - \$15,000 = \$85,000$ .



# Population variance and standard deviation

The **population variance** is the average of the squares of the distance each value is from the mean. The symbol for the population variance is  $\sigma^2$  ( $\sigma$  is the Greek lower-case letter sigma).

The formula for the population variance is

$$\sigma^2 = \frac{\sum(X - \mu)^2}{N}$$

where  $X$  = individual value

$\mu$  = population mean

$N$  = population size

The **population standard deviation** is the square root of the variance. The symbol for the population standard deviation is  $\sigma$ .

The corresponding formula for the population standard deviation is

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum(X - \mu)^2}{N}}$$



# Population variance and standard deviation

## Procedure Table

### Finding the Population Variance and Population Standard Deviation

**Step 1** Find the mean for the data.

$$\mu = \frac{\sum X}{N}$$

**Step 2** Find the deviation for each data value.

$$X - \mu$$

**Step 3** Square each of the deviations.

$$(X - \mu)^2$$

**Step 4** Find the sum of the squares.

$$\sum (X - \mu)^2$$

**Step 5** Divide by  $N$  to get the variance.

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}$$

**Step 6** Take the square root of the variance to get the standard deviation.

$$\sigma = \sqrt{\frac{\sum (X - \mu)^2}{N}}$$



## Example 23: Population variance and standard deviation

Find the variance and standard deviation for the number of months, given below, the given brand A lasted before fading.

10, 60, 50, 30, 40, 20

### SOLUTION

**Step 1** Find the mean for the data.

$$\mu = \frac{\sum X}{N} = \frac{10 + 60 + 50 + 30 + 40 + 20}{6} = \frac{210}{6} = 35$$

**Step 2** Subtract the mean from each data value ( $X - \mu$ ).

$$\begin{array}{lll} 10 - 35 = -25 & 50 - 35 = +15 & 40 - 35 = +5 \\ 60 - 35 = +25 & 30 - 35 = -5 & 20 - 35 = -15 \end{array}$$



## Example 23: Population variance and standard deviation

### SOLUTION

**Step 3** Square each result  $(X - \mu)^2$ .

$$\begin{array}{lll} (-25)^2 = 625 & (+15)^2 = 225 & (+5)^2 = 25 \\ (+25)^2 = 625 & (-5)^2 = 25 & (-15)^2 = 225 \end{array}$$

**Step 4** Find the sum of the squares  $\Sigma(X - \mu)^2$ .

$$625 + 625 + 225 + 25 + 25 + 225 = 1750$$

**Step 5** Divide the sum by  $N$  to get the variance  $\frac{[\Sigma(X - \mu)^2]}{N}$ .

$$\text{Variance} = 1750 \div 6 \approx 291.7$$



# Example 23: Population variance and standard deviation

## SOLUTION

**Step 6** Take the square root of the variance to get the standard deviation. Hence, the standard deviation equals  $\sqrt{291.7}$ , or 17.1. It is helpful to make a table.

| A<br>Values $X$ | B<br>$X - \mu$ | C<br>$(X - \mu)^2$ |
|-----------------|----------------|--------------------|
| 10              | -25            | 625                |
| 60              | +25            | 625                |
| 50              | +15            | 225                |
| 30              | -5             | 25                 |
| 40              | +5             | 25                 |
| 20              | -15            | 225                |
|                 |                | <hr/> 1750         |

Column A contains the raw data  $X$ . Column B contains the differences  $X - \mu$  obtained in step 2. Column C contains the squares of the differences obtained in step 3.



## Example 24: Population variance and standard deviation

Find the variance and standard deviation for the number of months, given below, the given brand B lasted before fading.

35, 45, 30, 35, 40, 25

### SOLUTION

**Step 1** Find the mean.

$$\mu = \frac{\Sigma X}{N} = \frac{35 + 45 + 30 + 35 + 40 + 25}{6} = \frac{210}{6} = 35$$

**Step 2** Subtract the mean from each value, and place the result in column B of the table.

|               |                |                 |
|---------------|----------------|-----------------|
| $35 - 35 = 0$ | $45 - 35 = 10$ | $30 - 35 = -5$  |
| $35 - 35 = 0$ | $40 - 35 = 5$  | $25 - 35 = -10$ |





## Example 24: Population variance and standard deviation

**SOLUTION** **Step 3** Square each result and place the squares in column C of the table.

| A<br>$X$ | B<br>$X - \mu$ | C<br>$(X - \mu)^2$ |
|----------|----------------|--------------------|
| 35       | 0              | 0                  |
| 45       | 10             | 100                |
| 30       | -5             | 25                 |
| 35       | 0              | 0                  |
| 40       | 5              | 25                 |
| 25       | -10            | 100                |

**Step 4** Find the sum of the squares in column C.

$$\Sigma(X - \mu)^2 = 0 + 100 + 25 + 0 + 25 + 100 = 250$$

**Step 5** Divide the sum by  $N$  to get the variance.

$$\sigma^2 = \frac{\Sigma(X - \mu)^2}{N} = \frac{250}{6} = 41.7$$



## Example 24: Population variance and standard deviation

### SOLUTION

**Step 6** Take the square root to get the standard deviation.

$$\sigma = \sqrt{\frac{\Sigma(X - \mu)^2}{N}} = \sqrt{41.7} \approx 6.5$$

Hence, the standard deviation is 6.5.

---

Since the standard deviation of brand A is 17.1 and the standard deviation of brand B is 6.5, the data are more variable for brand A. *In summary, when the means are equal, the larger the variance or standard deviation is, the more variable the data are.*



# Sample variance and standard deviation

## Formula for the Sample Variance

The formula for the sample variance (denoted by  $s^2$ ) is

$$s^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

where  $X$  = individual value

$\bar{X}$  = sample mean

$n$  = sample size

## Formula for the Sample Standard Deviation

The formula for the sample standard deviation, denoted by  $s$ , is

$$s = \sqrt{s^2} = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}}$$

where  $X$  = individual value

$\bar{X}$  = sample mean

$n$  = sample size



## Example 25: Sample variance and standard deviation

The number of public school teacher strikes in Pennsylvania for a random sample of school years is shown. Find the sample variance and the sample standard deviation.

9      10      14      7      8      3

### SOLUTION

**Step 1** Find the mean of the data values.

$$\bar{X} = \frac{\Sigma X}{n} = \frac{9 + 10 + 14 + 7 + 8 + 3}{6} = \frac{51}{6} = 8.5$$

**Step 2** Find the deviation for each data value ( $X - \bar{X}$ ).

$$\begin{array}{lll} 9 - 8.5 = 0.5 & 10 - 8.5 = 1.5 & 14 - 8.5 = 5.5 \\ 7 - 8.5 = -1.5 & 8 - 8.5 = -0.5 & 3 - 8.5 = -5.5 \end{array}$$



# Example 25: Sample variance and standard deviation

## SOLUTION

**Step 3** Square each of the deviations  $(X - \bar{X})^2$ .

$$\begin{array}{lll} (0.5)^2 = 0.25 & (1.5)^2 = 2.25 & (5.5)^2 = 30.25 \\ (-1.5)^2 = 2.25 & (-0.5)^2 = 0.25 & (-5.5)^2 = 30.25 \end{array}$$

**Step 4** Find the sum of the squares.

$$\Sigma(X - \bar{X})^2 = 0.25 + 2.25 + 30.25 + 2.25 + 0.25 + 30.25 = 65.5$$

**Step 5** Divide by  $n - 1$  to get the variance.

$$s^2 = \frac{\Sigma(X - \bar{X})^2}{n - 1} = \frac{65.5}{6 - 1} = \frac{65.5}{5} = 13.1$$

**Step 6** Take the square root of the variance to get the standard deviation.

$$s = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n - 1}} = \sqrt{13.1} \approx 3.6 \text{ (rounded)}$$

Here the sample variance is 13.1, and the sample standard deviation is 3.6.



# Sample variance and standard deviation

## Shortcut or Computational Formulas for $s^2$ and $s$

The shortcut formulas for computing the variance and standard deviation for data obtained from samples are as follows.

| Variance                                          | Standard deviation                                     |
|---------------------------------------------------|--------------------------------------------------------|
| $s^2 = \frac{n(\sum X^2) - (\sum X)^2}{n(n - 1)}$ | $s = \sqrt{\frac{n(\sum X^2) - (\sum X)^2}{n(n - 1)}}$ |

*Note that  $\sum X^2$  is not the same as  $(\sum X)^2$ . The notation  $\sum X^2$  means to square the values first, then sum;  $(\sum X)^2$  means to sum the values first, then square the sum.*



## Example 26: Sample variance and standard deviation

The number of public school teacher strikes in Pennsylvania for a random sample of school years is shown. Find the sample variance and sample standard deviation.

9, 10, 14, 7, 8, 3

### SOLUTION

**Step 1** Find the sum of the values:

$$\Sigma X = 9 + 10 + 14 + 7 + 8 + 3 = 51$$

**Step 2** Square each value and find the sum:

$$\Sigma X^2 = 9^2 + 10^2 + 14^2 + 7^2 + 8^2 + 3^2 = 499$$



# Example 26: Sample variance and standard deviation

## **SOLUTION**

**Step 3** Substitute in the formula and solve:

$$\begin{aligned}s^2 &= \frac{n(\sum X^2) - (\sum X)^2}{n(n - 1)} \\&= \frac{6(499) - 51^2}{6(6 - 1)} \\&= \frac{2994 - 2601}{6(5)} \\&= \frac{393}{30} \\&= 13.1\end{aligned}$$

The variance is 13.1.

$$s = \sqrt{13.1} \approx 3.6 \text{ (rounded)}$$

Hence, the sample variance is 13.1, and the sample standard deviation is 3.6.





# Variance and standard deviation for grouped data

## Shortcut or Computational Formula for $s^2$ and $s$ for Grouped Data

Sample variance:

$$s^2 = \frac{n(\sum f \cdot X_m^2) - (\sum f \cdot X_m)^2}{n(n - 1)}$$

Sample standard deviation

$$s = \sqrt{\frac{n(\sum f \cdot X_m^2) - (\sum f \cdot X_m)^2}{n(n - 1)}}$$

where  $X_m$  is the midpoint of each class and  $f$  is the frequency of each class.



# Variance and standard deviation for grouped data

## Procedure Table

### Finding the Sample Variance and Standard Deviation for Grouped Data

**Step 1** Make a table as shown, and find the midpoint of each class.

| A     | B         | C        | D             | E               |
|-------|-----------|----------|---------------|-----------------|
| Class | Frequency | Midpoint | $f \cdot X_m$ | $f \cdot X_m^2$ |

**Step 2** Multiply the frequency by the midpoint for each class, and place the products in column D.

**Step 3** Multiply the frequency by the square of the midpoint, and place the products in column E.

**Step 4** Find the sums of columns B, D, and E. (The sum of column B is  $n$ . The sum of column D is  $\Sigma f \cdot X_m$ . The sum of column E is  $\Sigma f \cdot X_m^2$ .)

**Step 5** Substitute in the formula and solve to get the variance.

$$s^2 = \frac{n(\Sigma f \cdot X_m^2) - (\Sigma f \cdot X_m)^2}{n(n - 1)}$$

**Step 6** Take the square root to get the standard deviation.



## Example 27: Variance and standard deviation for grouped data

Find the sample variance and standard deviation for the frequency distribution of the data below, which represent the number of miles that 20 runners ran during one week.

| Class     | Frequency | Midpoint |
|-----------|-----------|----------|
| 5.5–10.5  | 1         | 8        |
| 10.5–15.5 | 2         | 13       |
| 15.5–20.5 | 3         | 18       |
| 20.5–25.5 | 5         | 23       |
| 25.5–30.5 | 4         | 28       |
| 30.5–35.5 | 3         | 33       |
| 35.5–40.5 | 2         | 38       |



## Example 27: Variance and standard deviation for grouped data

### SOLUTION

**Step 1** Make a table as shown, and find the midpoint of each class.

| <b>A</b><br><b>Class</b> | <b>B</b><br><b>Frequency</b><br>$f$ | <b>C</b><br><b>Midpoint</b><br>$X_m$ | <b>D</b><br>$f \cdot X_m$ | <b>E</b><br>$f \cdot X_m^2$ |
|--------------------------|-------------------------------------|--------------------------------------|---------------------------|-----------------------------|
| 5.5–10.5                 | 1                                   | 8                                    |                           |                             |
| 10.5–15.5                | 2                                   | 13                                   |                           |                             |
| 15.5–20.5                | 3                                   | 18                                   |                           |                             |
| 20.5–25.5                | 5                                   | 23                                   |                           |                             |
| 25.5–30.5                | 4                                   | 28                                   |                           |                             |
| 30.5–35.5                | 3                                   | 33                                   |                           |                             |
| 35.5–40.5                | 2                                   | 38                                   |                           |                             |



# Example 27: Variance and standard deviation for grouped data

## SOLUTION

**Step 2** Multiply the frequency by the midpoint for each class, and place the products in column D.

$$1 \cdot 8 = 8 \quad 2 \cdot 13 = 26 \quad \dots \quad 2 \cdot 38 = 76$$

**Step 3** Multiply the frequency by the square of the midpoint, and place the products in column E.

$$1 \cdot 8^2 = 64 \quad 2 \cdot 13^2 = 338 \quad \dots \quad 2 \cdot 38^2 = 2888$$

**Step 4** Find the sums of columns B, D, and E. The sum of column B is  $n$ , the sum of column D is  $\Sigma f \cdot X_m$ , and the sum of column E is  $\Sigma f \cdot X_m^2$ . The completed table is shown.

| A<br>Class | B<br>Frequency | C<br>Midpoint | D<br>$f \cdot X_m$         | E<br>$f \cdot X_m^2$            |
|------------|----------------|---------------|----------------------------|---------------------------------|
| 5.5–10.5   | 1              | 8             | 8                          | 64                              |
| 10.5–15.5  | 2              | 13            | 26                         | 338                             |
| 15.5–20.5  | 3              | 18            | 54                         | 972                             |
| 20.5–25.5  | 5              | 23            | 115                        | 2,645                           |
| 25.5–30.5  | 4              | 28            | 112                        | 3,136                           |
| 30.5–35.5  | 3              | 33            | 99                         | 3,267                           |
| 35.5–40.5  | 2              | 38            | 76                         | 2,888                           |
|            | $n = 20$       |               | $\Sigma f \cdot X_m = 490$ | $\Sigma f \cdot X_m^2 = 13,310$ |



# Example 27: Variance and standard deviation for grouped data

## SOLUTION

**Step 5** Substitute in the formula and solve for  $s^2$  to get the variance.

$$\begin{aligned}s^2 &= \frac{n(\Sigma f \cdot X_m^2) - (\Sigma f \cdot X_m)^2}{n(n - 1)} \\&= \frac{20(13,310) - 490^2}{20(20 - 1)} \\&= \frac{266,200 - 240,100}{20(19)} \\&= \frac{26,100}{380} \\&\approx 68.7\end{aligned}$$

**Step 6** Take the square root to get the standard deviation.

$$s \approx \sqrt{68.7} \approx 8.3$$



# Summary of measures of variation

| Summary of Measures of Variation |                                                                         |                 |
|----------------------------------|-------------------------------------------------------------------------|-----------------|
| Measure                          | Definition                                                              | Symbol(s)       |
| Range                            | Distance between highest value and lowest value                         | $R$             |
| Variance                         | Average of the squares of the distance that each value is from the mean | $\sigma^2, s^2$ |
| Standard deviation               | Square root of the variance                                             | $\sigma, s$     |



## Example 28: The coefficient of variation

The **coefficient of variation**, denoted by CVar, is the standard deviation divided by the mean. The result is expressed as a percentage.

**For samples,**

$$\text{CVar} = \frac{s}{\bar{X}} \cdot 100$$

**For populations,**

$$\text{CVar} = \frac{\sigma}{\mu} \cdot 100$$

The mean of the number of sales of cars over a 3-month period is 87, and the standard deviation is 5. The mean of the commissions is \$5225, and the standard deviation is \$773. Compare the variations of the two.

### SOLUTION

The coefficients of variation are

$$\text{CVar} = \frac{s}{\bar{X}} = \frac{5}{87} \cdot 100 = 5.7\% \quad \text{sales}$$

$$\text{CVar} = \frac{773}{5225} \cdot 100 = 14.8\% \quad \text{commissions}$$

Since the coefficient of variation is larger for commissions, the commissions are more variable than the sales.





# The Chebyshev's theorem

*Chebyshev's theorem*, developed by the Russian mathematician Chebyshev (1821–1894), specifies the proportions of the spread in terms of the standard deviation.

**Chebyshev's theorem** The proportion of values from a data set that will fall within  $k$  standard deviations of the mean will be at least  $1 - 1/k^2$ , where  $k$  is a number greater than 1 ( $k$  is not necessarily an integer).

This theorem states that at least three-fourths, or 75%, of the data values will fall within 2 standard deviations of the mean of the data set. This result is found by substituting  $k = 2$  in the expression

$$1 - \frac{1}{k^2} \quad \text{or} \quad 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4} = 75\%$$



# The Chebyshev's theorem

Furthermore, the theorem states that at least eight-ninths, or 88.89%, of the data values will fall within 3 standard deviations of the mean. This result is found by letting  $k = 3$  and substituting in the expression.

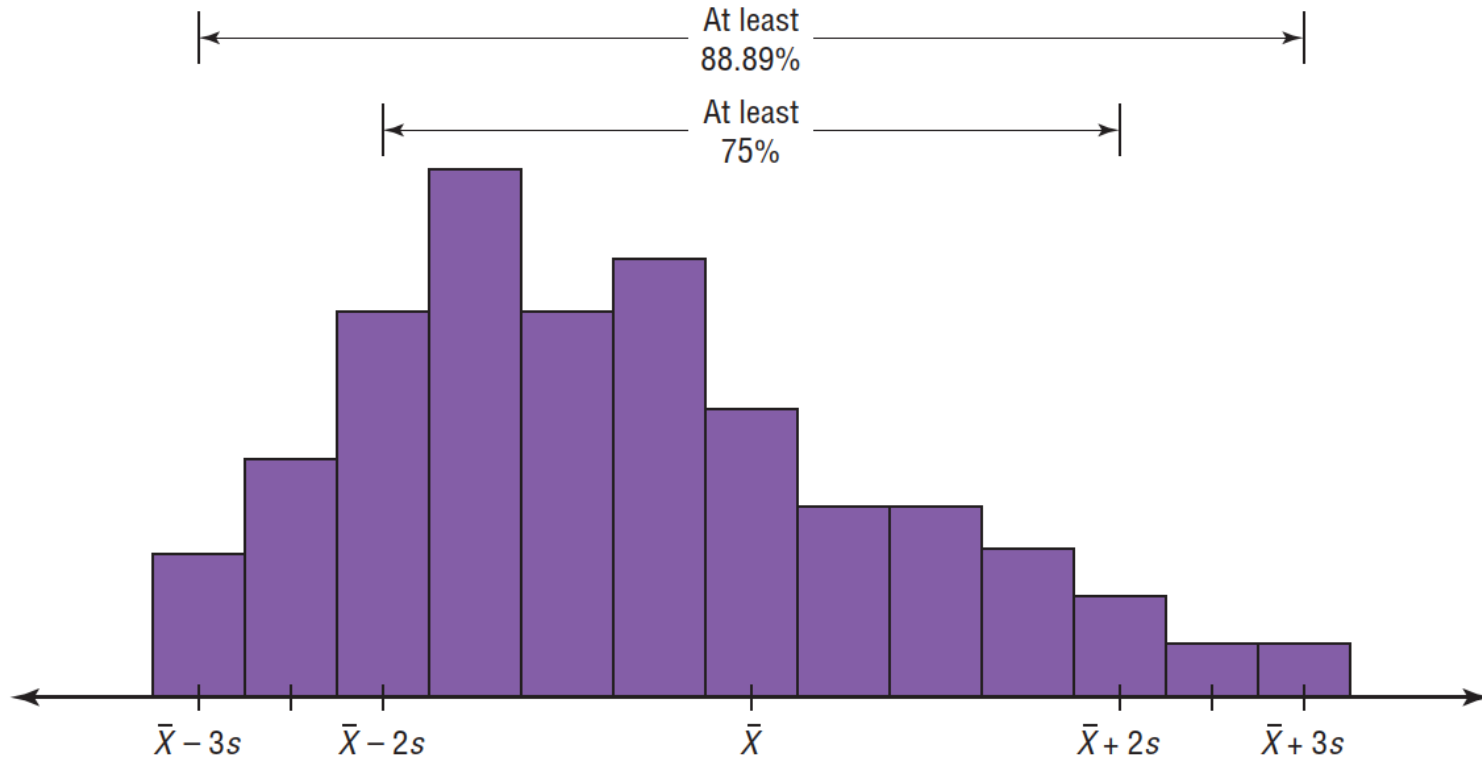
$$1 - \frac{1}{k^2} \quad \text{or} \quad 1 - \frac{1}{3^2} = 1 - \frac{1}{9} = \frac{8}{9} = 88.89\%$$

In summary, then, Chebyshev's theorem states

- At least three-fourths, or 75%, of all data values fall within 2 standard deviations of the mean.
- At least eight-ninths, or 89%, of all data values fall within 3 standard deviations of the mean.



# The Chebyshev's theorem



This theorem can be applied to any distribution regardless of its shape.



## Example 29: The Chebyshev's theorem

The mean price of houses in a certain neighborhood is \$50,000, and the standard deviation is \$10,000. Find the price range for which at least 75% of the houses will sell.

### SOLUTION

Chebyshev's theorem states that three-fourths, or 75%, of the data values will fall within 2 standard deviations of the mean. Thus,

$$\$50,000 + 2(\$10,000) = \$50,000 + \$20,000 = \$70,000$$

and

$$\$50,000 - 2(\$10,000) = \$50,000 - \$20,000 = \$30,000$$

Hence, at least 75% of all homes sold in the area will have a price range from \$30,000 to \$70,000.



## Example 30: The Chebyshev's theorem

A survey of local companies found that the mean amount of travel allowance for couriers was \$0.25 per mile. The standard deviation was \$0.02. Using Chebyshev's theorem, find the minimum percentage of the data values that will fall between \$0.20 and \$0.30.

### SOLUTION

**Step 1** Subtract the mean from the larger value.

$$\$0.30 - \$0.25 = \$0.05$$

**Step 2** Divide the difference by the standard deviation to get  $k$ .

$$k = \frac{0.05}{0.02} = 2.5$$

**Step 3** Use Chebyshev's theorem to find the percentage.

$$1 - \frac{1}{k^2} = 1 - \frac{1}{2.5^2} = 1 - \frac{1}{6.25} = 1 - 0.16 = 0.84 \quad \text{or} \quad 84\%$$

Hence, at least 84% of the data values will fall between \$0.20 and \$0.30.



# The empirical (normal) rule

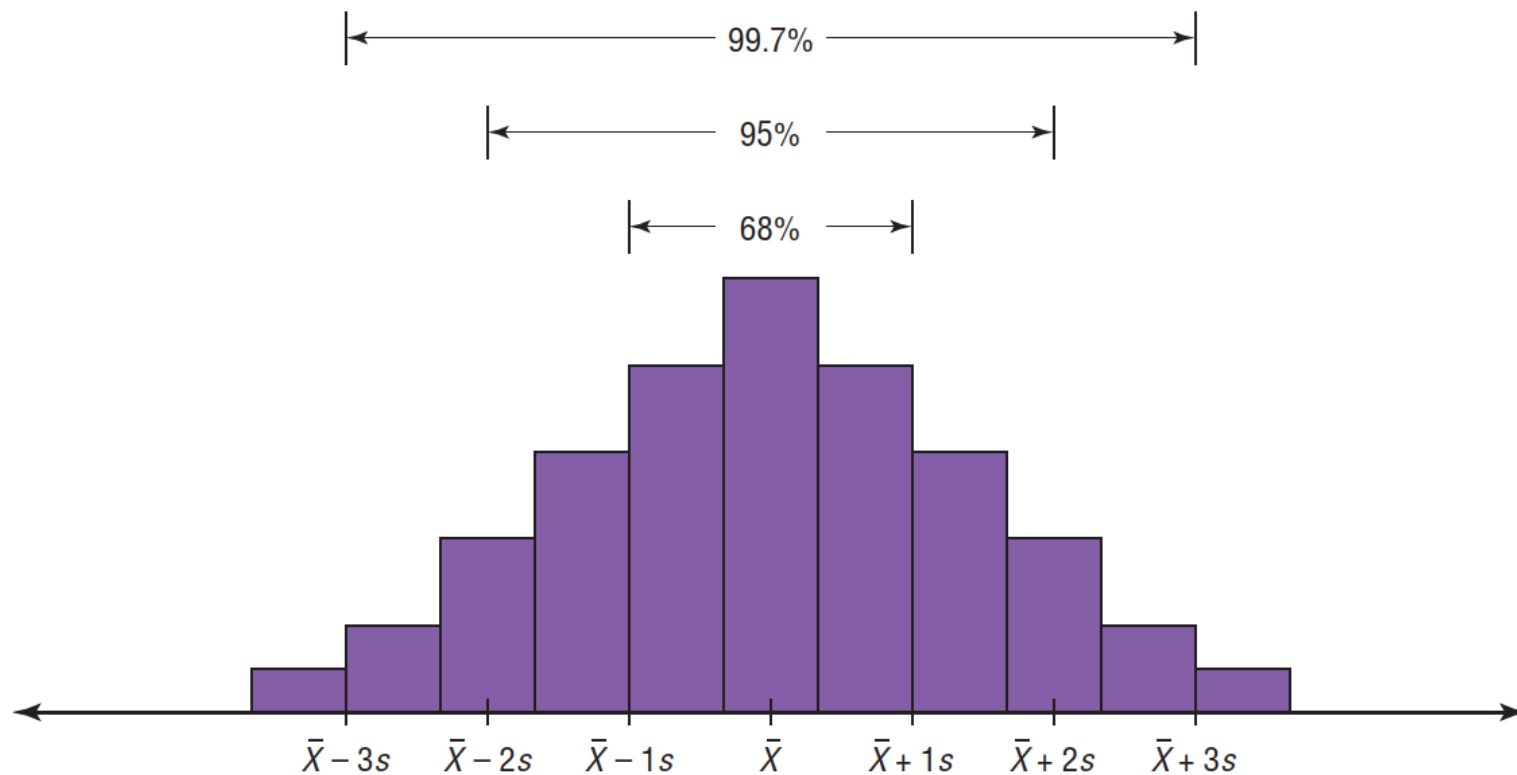
## The Empirical (Normal) Rule

Chebyshev's theorem applies to any distribution regardless of its shape. However, when a distribution is *bell-shaped* (or what is called *normal*), the following statements, which make up the **empirical rule**, are true.

- Approximately 68% of the data values will fall within 1 standard deviation of the mean.
- Approximately 95% of the data values will fall within 2 standard deviations of the mean.
- Approximately 99.7% of the data values will fall within 3 standard deviations of the mean.



# The empirical (normal) rule



# The empirical (normal) rule

For example, suppose that the scores on a national achievement exam have a mean of 480 and a standard deviation of 90. If these scores are normally distributed, then approximately 68% will fall between 390 and 570 ( $480 + 90 = 570$  and  $480 - 90 = 390$ ). Approximately 95% of the scores will fall between 300 and 660 ( $480 + 2 \cdot 90 = 660$  and  $480 - 2 \cdot 90 = 300$ ). Approximately 99.7% will fall between 210 and 750 ( $480 + 3 \cdot 90 = 750$  and  $480 - 3 \cdot 90 = 210$ ).

Because the empirical rule requires that the distribution be approximately bell-shaped, the results are more accurate than those of Chebyshev's theorem, which applies to all distributions.

